

303Modbus13 RTU Control — User Guide

Step-by-step instructions for the desktop application that controls the **Ideaflex 303Modbus13** relay module over **Modbus RTU**.

For register addresses, timeouts, and wiring details see [REGISTERS.md](#).

Use the screenshots in each section to find the matching buttons and fields in the application window.

Language

In the top-right corner of the window, use the **Language** dropdown to switch between **English** and **Polish**. The choice is saved in `settings.json` for the next launch. For documentation screenshots, select **English**.

1. Before you start

Item	Typical value
Connection	USB-RS485 converter (e.g. CH340)
COM port	e.g. COM4 (check Device Manager)
Baud rate	9600, format 8N1
Factory Slave ID	FF (255 decimal)
Relays	CH1, CH2 (coils 0-1)
Opto inputs	IN1, IN2 (discrete inputs 0-1)

Wire the module: RS-485 **A to A**, **B to B**, power supply. For opto inputs connect **COM to signal GND** and **INx to +12-24 V** (see [REGISTERS.md](#)).

Run the application:

```
python app.py
```

2. Connection tab

Before connecting (select COM port and Slave ID):

303Modbus13 — Modbus RTU Control

303Modbus132-channel relay module — Modbus RTU control

DISCONNECTEDno connection to module

Language: English

ConnectionControlAdvancedAbout

Serial port (Modbus RTU)

COM port:COM4

Baud rate:9600

Slave ID (hex):01

ConnectSearch for module

Tip: factory address is FF (255). After setting a new address, connect using that address (e.g. 01).

STATUS: DISCONNECTED

Click Connect to start communication with the module.

Change Modbus address (broadcast)

Write to register 0 with slave=0 (broadcast). After change, disconnect and connect with the new address.

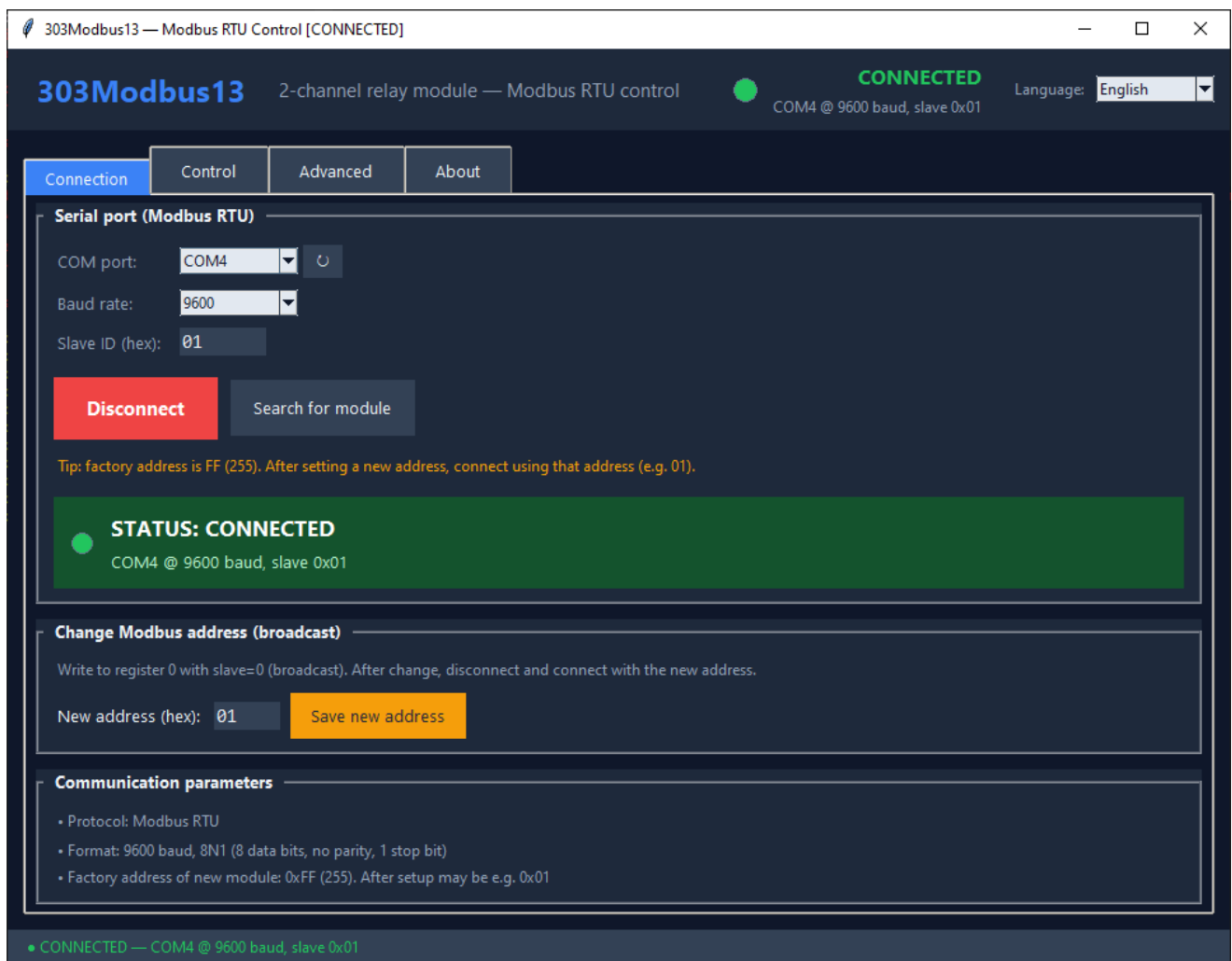
New address (hex):01Save new address

Communication parameters

- Protocol: Modbus RTU
- Format: 9600 baud, 8N1 (8 data bits, no parity, 1 stop bit)
- Factory address of new module: 0xFF (255). After setup may be e.g. 0x01

DISCONNECTED — select COM port and click Connect

After a successful connection (green status banner):



Connect to the module

1. Plug in the USB-RS485 adapter and note the COM port in **Device Manager**.
2. Select **Port COM** (e.g. COM4). Click the refresh button if the port is not listed.
3. Set **Baud rate** to 9600.
4. Set **Slave ID (hex)** to FF for a new module (factory address).
5. Click **Connect**.

When connected, the status bar and banner turn green (**CONNECTED**) with details such as COM4 @ 9600 baud, slave 0xFF.

Find module automatically

Click **Search for module**. The app scans COM ports, baud rates, and common Slave IDs. If found, port and Slave ID are filled in automatically.

Change Modbus address

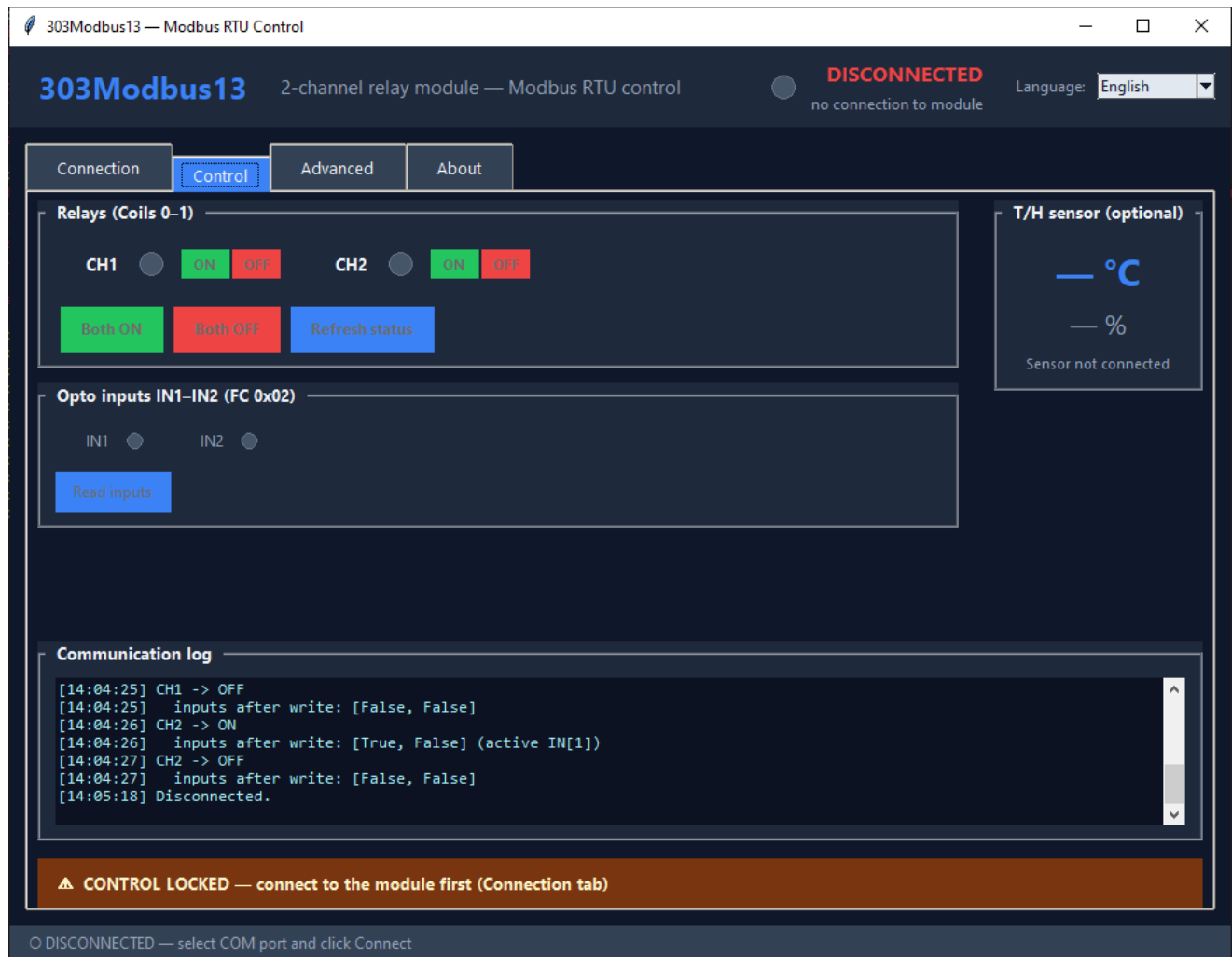
Use this only with **one module** on the bus:

1. Stay connected with the current address (usually FF).

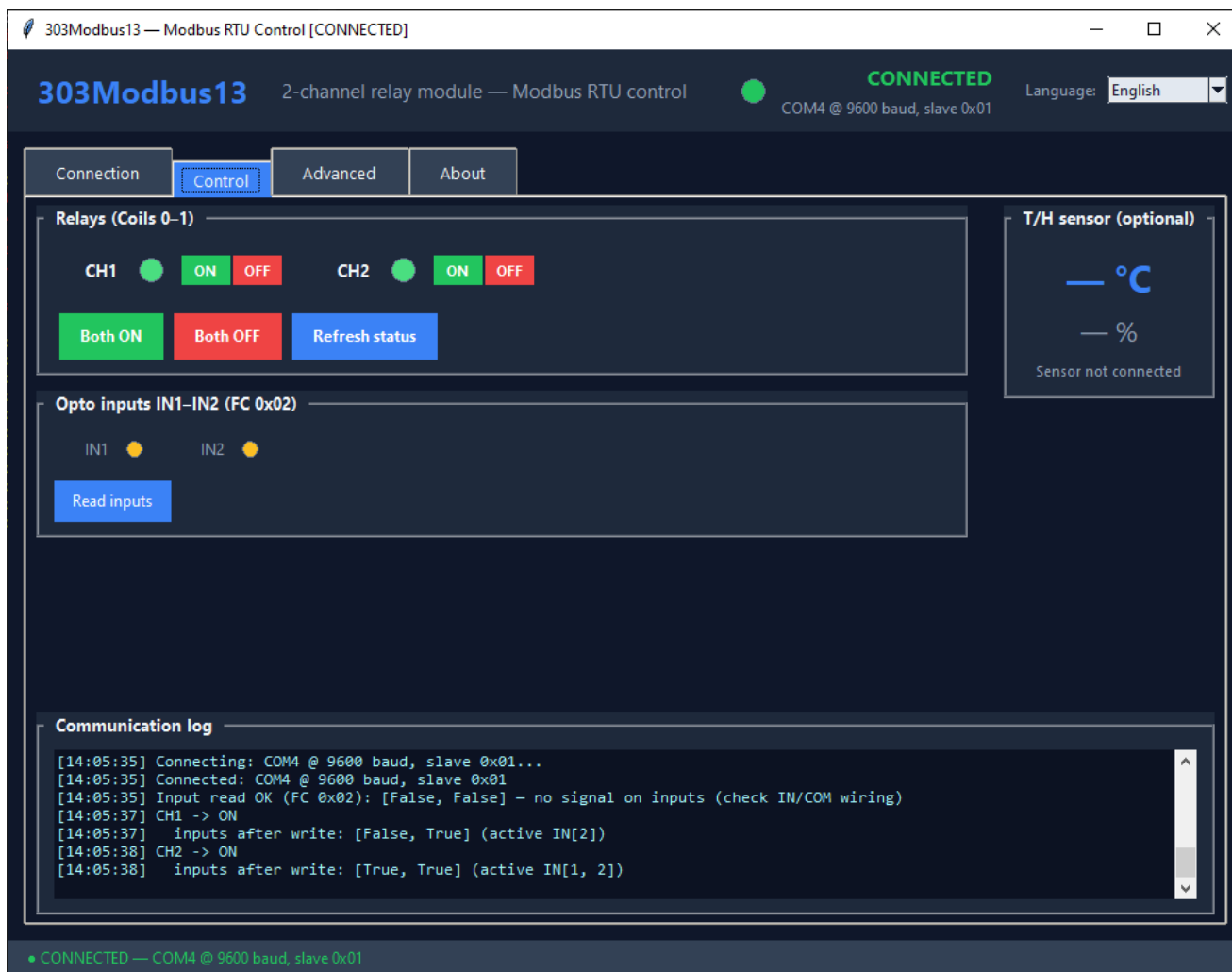
2. In **New address (hex)** enter the new ID, e.g. 01.
3. Click **Save new address** — this sends FC 0x10 with Slave = 0 (broadcast).
4. **Disconnect**, set Slave ID to the new value (01), and **connect** again.

3. Control tab

When disconnected, controls are locked until you connect on the Connection tab:



When connected, relays and inputs are active. The log shows commands and automatic input reads after each coil write:



Relays

Control	Action
ON / OFF per channel	Toggle CH1 or CH2
Both ON	Turn both relays on (FC 0x0F, 8 coils in frame)
Both OFF	Turn both relays off
Refresh status	Refresh relay, input, and optional T/H sensor state

The LED next to each channel is **green** when the relay is ON, **grey** when OFF.

After each relay operation the app reads opto inputs automatically and shows the result in the log.

Opto inputs

Element	Meaning
IN1 / IN2 LED	Yellow = active input, grey = inactive

Read inputs	Manual read (FC 0x02)
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Example log line when IN2 is active:

Input read OK (FC 0x02): [False, True] - active: IN2

Optional T/H sensor

If a temperature/humidity sensor is installed (Slave ID 02), values appear in the right panel. Otherwise you will see **Sensor not connected** — this is normal.

Input LEDs update automatically after each relay command (ON/OFF, Both ON/OFF). Use **Refresh** for a manual full readout, or **Read inputs** for inputs only.

Communication log

The bottom **Communication log** panel records connections, relay commands, input reads, and errors. Use it for quick diagnosis.

4. Advanced tab

The screenshot shows the '303Modbus13' application window. The title bar reads '303Modbus13 — Modbus RTU Control [CONNECTED]'. The interface has a dark theme. At the top, it says '2-channel relay module — Modbus RTU control' and 'CONNECTED' in green. Below this, it shows 'COM4 @ 9600 baud, slave 0x01' and a language dropdown set to 'English'. The main menu has four tabs: 'Connection', 'Control', 'Advanced' (which is selected and highlighted with a blue border), and 'About'. The 'Advanced' tab contains the 'Modbus console — any command' section. It includes instructions: 'Select a Modbus function, address, count, and values (comma-separated). Read coils/inputs: count=8. Write multiple coils (0x0F): values 1,1,1,1,1,1,1 or single 1/0.' The form fields are: 'Function:' with a dropdown menu showing 'Read Coils (0x01)', 'Address:' with a text input '0', 'Count:' with a text input '8', 'Values:' with an empty text input, and 'Slave (hex, empty=current):' with an empty text input. Below these is a blue 'Execute' button. Under the 'Execute' button, the 'Result:' is displayed in a light green box: '[True, True, False, False, False, False, False, False]'. At the bottom of the window, a status bar shows '• CONNECTED — COM4 @ 9600 baud, slave 0x01'.

The **Modbus console** sends raw Modbus frames for testing and integration work.

Field	Description
Function	Modbus function (FC 0x01-0x06, 0x0F, 0x10)
Address	Start address
Count	Number of items (for reads / write_coils length)
Values	Comma-separated values for write functions (0 / 1 for coils)
Slave	Hex Slave ID; leave empty to use the connected address
Execute	Run the command
Result	Response or error message

Example commands

Goal	Function	Address	Count	Values	Slave
Read relays	Read Coils (0x01)	0	8	-	empty
Read inputs	Read Discrete Inputs (0x02)	0	8	-	empty
CH1 ON	Write Single Coil (0x05)	0	1	1	empty
Both ON	Write Multiple Coils (0x0F)	0	8	1,1,1,1,1,1,1,1	empty
Set address 1	Write Multiple Registers (0x10)	0	1	1	0
CH1 ON 3 s, then auto OFF	Write Multiple Registers (0x10)	3	2	4,30	empty

Delayed relay switching (verified on hardware)

The module supports timed pulses in addition to immediate ON/OFF. Use **FC 0x10** on holding-register blocks — **not** the coil addresses used by the Control tab.

303Modbus13 — Modbus RTU Control [CONNECTED]

303Modbus13

2-channel relay module — Modbus RTU control

CONNECTED

COM4 @ 9600 baud, slave 0x01

Language: English

Connection

Control

Advanced

About

Modbus console — any command

Select a Modbus function, address, count, and values (comma-separated). Read coils/inputs: count=8. Write multiple coils (0x0F): values 1,1,1,1,1,1,1 or single 1/0.

Function:

Write Multiple Registers (0x10)

Address:

3

Count:

2

Values:

4, 30

Slave (hex, empty=current):

Execute

Result:

OK — rejestry od 3, count=2

CONNECTED — COM4 @ 9600 baud, slave 0x01

Screenshot above: connected at Slave 01, command **Write Multiple Registers (0x10)**, Address 3, Count 2, Values 4, 30.

Effect on CH1: relay turns **ON immediately**, then turns **OFF automatically after 3.0 s** ($30 \times 0.1 \text{ s}$). If NO1 is wired to IN1, you will see IN1 go active while the relay is on.

Field	CH1 (3 s pulse ON)	CH2 (3 s pulse ON)
Function	Write Multiple Registers (0x10)	same
Address	3	8
Count	2	2
Values	4, 30	4, 30
Slave	empty (uses connected ID)	same

Delay modes (first value in values; second value = time in **0.1 s** units, range 1–65535):

Mode	Value	Behaviour
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Pulse ON	4	Relay ON → auto OFF after delay
Pulse OFF	2	Relay OFF → auto ON after delay
Manual	—	Use FC 0x05 / 0x0F on coil addresses 0 / 1 (Control tab) — immediate, no timer

Only modes **2** and **4** are used for delayed switching. There are no other documented timer modes on this module.

More examples:

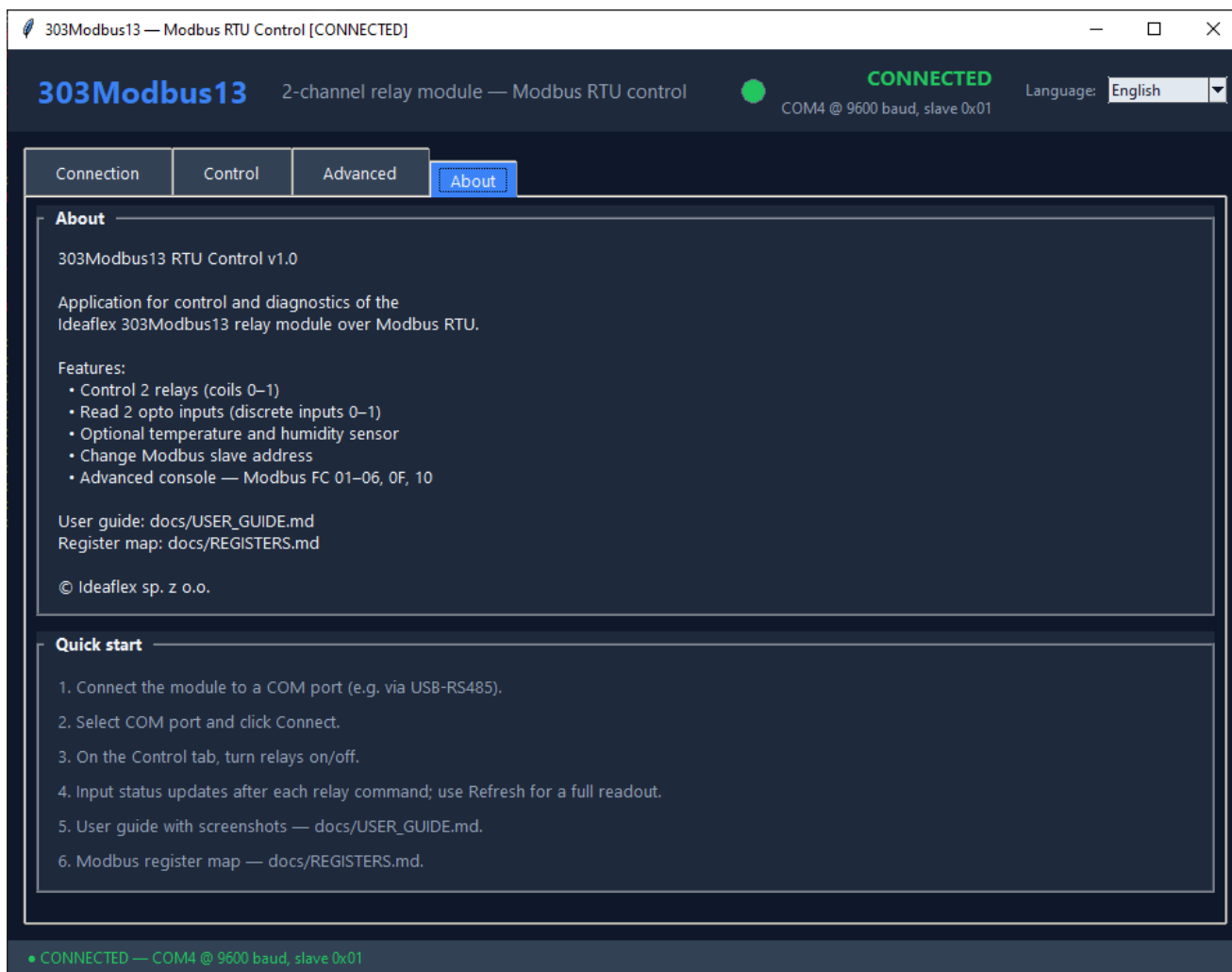
Goal	Address	Values	Time
CH1 ON 2 s, then OFF	3	4, 20	2.0 s
CH1 OFF 5 s, then ON	3	2, 50	5.0 s
CH2 ON 1 s, then OFF	8	4, 10	1.0 s

Full register map: [REGISTERS.md](#) (section *Delayed switching*).

Tips:

- Use **count = 8** for reading coils and discrete inputs on this module.
- For **Write Multiple Coils**, you can enter a single value **1** or **0** — it will be expanded to 8 coils.
- Wait until any other operation finishes before clicking **Execute** (or click **Cancel** on the Connection tab).

5. About tab



Shows application version, feature list, and a short quick-start checklist. Full documentation:

Document	Content
USER_GUIDE.md	This guide (screenshots and UI instructions)
REGISTERS.md	Modbus register map, timeouts, wiring
README.md	Installation and project overview

6. Quick troubleshooting

Symptom	What to check
Cannot connect	Correct COM port, cable A/B, Slave ID FF or 01, close other apps using the port
Timeout	Use 9600 8N1; module needs ~3 s timeout (set in app backend)

Relays work, inputs always off	Wire COM to GND and INx to +signal on the module (not only on relay contacts)
Both ON/OFF does nothing	Ensure you are connected; check log for errors
Console does nothing	Another operation may be in progress — wait or cancel
T/H sensor missing	Optional hardware — ignore if not installed

7. Contact

If you have any problems or questions, contact **FlexTech_KRK** on [Allegro](#) or email biuro@ideaflex.pl.

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